UNIVERSITY OF WATERLOO

SYDE 556/750 – Simulating Neurobiological Systems

Test 1: Neurons & Population Representation

Student ID: _____ Time: 30 minutes Total: 20 marks (plus 2 bonus)

- **Closed book.** No notes, textbook, calculator, phone, or other devices.
- The marks for each question are shown in brackets on the right.
- Write your answers in the space provided. Show your work — partial credit is given for correct reasoning.

Question 1. **[4 marks]**

A rectified-linear neuron fires at rate $a = G[J] = \max(J, 0)$, with input current $J = \alpha\langle x, e\rangle + J^{\text{bias}}$. Consider a one-dimensional neuron with encoder $e = +1$. Derive expressions for the gain α and the bias J^{bias} so that the neuron has maximum firing rate $a^{\max}$ at $x = 1$ and an x-intercept at $x = \xi$ (the value of x at which it begins to fire). Show your work.

Question 2. **[4 marks]**

In the box below, sketch firing rate (y-axis) versus $x \in [-1, 1]$ (x-axis) for three LIF neurons that all share the same maximum firing rate $a^{\max}$. Label each curve and mark its x-intercept.

(i) encoder $e = +1$, x-intercept near -0.5;

(ii) encoder $e = -1$, x-intercept near 0;

(iii) encoder $e = +1$, x-intercept near $+0.8$.

Question 3. **[4 marks]**

A population of N neurons represents a scalar value x. The input is sampled at M values, giving the population activity matrix $\mathbf{A}$.

(a) Write the matrix equation for the optimal decoders $\mathbf{d}$ that reconstruct x from $\mathbf{A}$, *including* the term that accounts for neural noise of variance σ^2.

(b) Now suppose the population represents a 2-D vector $\mathbf{x}$ (with $N = 10$ neurons) and at some moment has produced the activity vector $\mathbf{a}$. How do you calculate the decoded estimate $\hat{\mathbf{x}}$? Write the equation, and draw each vector and matrix in it as a labelled rectangle with its dimensions.

(c) In one or two sentences, explain why including the noise term changes how the decoders behave when the activities are noisy.

Question 4. **[4 marks]**

The representation error of a neural population has two distinct sources. (a) Name both. (b) State how each scales with the number of neurons n. (c) In one sentence, explain why they scale differently.

Question 5. **[4 marks]**

[Page 3 of the original paper was not photographed. Transcribe Question 5 here.]

Bonus Question 1. **[2 bonus marks]**

[Not photographed. Transcribe the bonus question here.]